Assignment-8: Function and Class Templates

Mansha Bhatnagar
1024150192
2O23

Function Template

1. Write a function template to swap two variables of any data type.

D: > COLLEGE > oops > ass7 > C++ ass8Q1.cpp

```cpp
#include <iostream>
using namespace std;

template <typename T>
void swapVars(T &a, T &b) {
    T temp = a;
    a = b;
    b = temp;
}

int main() {
    int x = 10, y = 20;
    swapVars(x, y);
    cout << x << " " << y << endl;

    double m = 1.5, n = 4.5;
    swapVars(m, n);
    cout << m << " " << n << endl;
    return 0;
}
```

```
PS C:\Users\HP> cd "d:\COLLEGE\o
s8Q1 }
20 10
4.5 1.5
PS D:\COLLEGE\oops\ass7>
```

2. Write a function template to find the minimum element in an array

D: > COLLEGE > oops > ass7 > ASS8Q2.cpp > main()

```cpp
#include <iostream>
using namespace std;

template <typename T>
T findMin(T arr[], int n) {
    T minVal = arr[0];
    for (int i = 1; i < n; i++) {
        if (arr[i] < minVal)
            minVal = arr[i];
    }
    return minVal;
}

int main() {
    int a[] = {5, 2, 8, 1, 9};
    cout << "Min int: " << findMin(a, 5) << endl;

    float b[] = {2.5, 0.5, 3.2};
    cout << "Min float: " << findMin(b, 3) << endl;
    return 0;
}
```

```
PS C:\Users\HP> cd "d:\CO
Q1 } ; if ($?) { .\ass8Q1
20 10
4.5 1.5
PS D:\COLLEGE\oops\ass7>
p -o ASS8Q2 } ; if ($?) {
Min int: 1
Min float: 0.5
PS D:\COLLEGE\oops\ass7>
```

3. Write a function template to sort an array using bubble sort.

D: > COLLEGE > oops > ass7 > ass8Q3.cpp > main()

```cpp
#include <iostream>
using namespace std;

template <typename T>
void bubbleSort(T arr[], int n) {
    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (arr[j] > arr[j + 1]) {
                T temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }
}

int main() {
    int arr[] = {64, 34, 25, 12, 22};
    bubbleSort(arr, 5);
    for (int i = 0; i < 5; i++) cout << arr[i] << " ";
    return 0;
}
```

```
PS C:\Users\HP> cd "d:\
Q3 } ; if ($?) { .\ass8
12 22 25 34 64
PS D:\COLLEGE\oops\ass7>
```

4. Write a function template to perform linear search.

```cpp
#include <iostream>
using namespace std;

template <typename T>
int linearSearch(T arr[], int n, T key) {
    for (int i = 0; i < n; i++) {
        if (arr[i] == key) return i;
    }
    return -1;
}

int main() {
    int arr[] = {10, 20, 30, 40};
    int pos = linearSearch(arr, 4, 30);
    if (pos != -1) cout << "Found at index: " << pos;
    else cout << "Not found";
    return 0;
}
```

```
PS C:\Users\HP> cd "
Q3 } ; if ($?) { .\a
12 22 25 34 64
PS D:\COLLEGE\oops\a
 tempCodeRunnerFile.
ile }
Found at index: 2
PS C:\Users\HP\AppDa
```

5. Design overloaded versions of a function template process() such that:

   i)Accepts a single parameter

   ii)Accepts two parameters of the same type

   iii) Accepts two parameters of different types

```cpp
#include <iostream>
using namespace std;

template <typename T>
void process(T a) {
    cout << "One parameter: " << a << endl;
}

template <typename T>
void process(T a, T b) {
    cout << "Two same types: " << a << ", " << b << endl;
}

template <typename T1, typename T2>
void process(T1 a, T2 b) {
    cout << "Two different types: " << a << ", " << b << endl;
}

int main() {
    process(10);
    process(10, 20);
    process(10, 5.5);
    return 0;
}
```

```
PS C:\Users\HP> cd "d:\COLLEGE\oops
Q3 } ; if ($?) { .\ass8Q3 }
12 22 25 34 64
PS D:\COLLEGE\oops\ass7> cd "C:\Use
 tempCodeRunnerFile.cpp -o tempCode
ile }
Found at index: 2
PS C:\Users\HP\AppData\Local\Temp>
($?) { g++ tempCodeRunnerFile.cpp -
odeRunnerFile }
One parameter: 10
Two same types: 10, 20
Two different types: 10, 5.5
PS C:\Users\HP\AppData\Local\Temp>
```

Class Template

1. Develop a class template for stack with push and pop operations.

ASS8Q4.cpp | ASS8Q5.cpp | #include <iostream> Untitled-7

```cpp
#include <iostream>
using namespace std;

template <class T>
class Stack {
    T arr[100];
    int top;
public:
    Stack() { top = -1; }
    void push(T val) {
        if (top < 99) arr[++top] = val;
    }
    void pop() {
        if (top >= 0) cout << "Popped: " << arr[top--] << endl;
    }
};

int main() {
    Stack<int> s;
    s.push(10);
    s.push(20);
    s.pop();
    return 0;
}
```

CHAT TERMINAL

```
PS C:\Users\HP> cd "C
++ tempCodeRunnerFile
mpCodeRunnerFile }
Popped: 20
PS C:\Users\HP\AppDat
```

2. Develop a class template for queue with enqueue and dequeue operations.

```cpp
#include <iostream>
using namespace std;

template <class T>
class Queue {
    T arr[100];
    int front, rear;
public:
    Queue() { front = 0; rear = 0; }
    void enqueue(T val) {
        arr[rear++] = val;
    }
    void dequeue() {
        if (front < rear) cout << "Dequeued: " << arr[front++] << endl;
    }
};

int main() {
    Queue<string> q;
    q.enqueue("Hello");
    q.enqueue("World");
    q.dequeue();
    return 0;
}
```

```
PS C:\Users\HP> cd
++ tempCodeRunnerFi
mpCodeRunnerFile }
Popped: 20
PS C:\Users\HP\AppD
emp\" ; if ($?) { g
 } ; if ($?) { .\te
Dequeued: Hello
PS C:\Users\HP\AppD
```

3. Create a class template to store and display a pair of values.

D: > COLLEGE > oops > ass7 > ASS8CC.cpp > main()

```cpp
#include <iostream>
using namespace std;

template <class T1, class T2>
class Pair {
    T1 first;
    T2 second;
public:
    Pair(T1 a, T2 b) : first(a), second(b) {}
    void display() {
        cout << "Pair: " << first << ", " << second << endl;
    }
};

int main() {
    Pair<string, int> p("Age", 20);
    p.display();
    return 0;
}
```

```
PS C:\Users\HP> cd "d:\CO
cpp -o ASS8CC } ; if ($?)
Pair: Age, 20
PS D:\COLLEGE\oops\ass7>
```

4. Create a class template for basic arithmetic operations.

D: > COLLEGE > oops > ass7 > ASS8DD.cpp > main()

```cpp
#include <iostream>
using namespace std;

template <class T>
class Calculator {
public:
    T add(T a, T b) { return a + b; }
    T sub(T a, T b) { return a - b; }
    T mul(T a, T b) { return a * b; }
    T div(T a, T b) { return a / b; }
};

int main() {
    Calculator<float> calc;
    cout << "Add: " << calc.add(10.5, 5.5) << endl;
    cout << "Div: " << calc.div(10.0, 2.0) << endl;
    return 0;
}
```

```
PS C:\Users\HP> cd "d:\COL
cpp -o ASS8CC } ; if ($?)
Pair: Age, 20
PS D:\COLLEGE\oops\ass7> c
+ ASS8DD.cpp -o ASS8DD } ;
Add: 16
Div: 5
PS D:\COLLEGE\oops\ass7>
```

5. Create a class template to input and display array elements

D: > COLLEGE > oops > ass7 > ASS8EE.cpp > main()

```cpp
#include <iostream>
using namespace std;

template <class T>
class ArrayHandler {
    T arr[10];
    int n;
public:
    void input(int size) {
        n = size;
        for (int i = 0; i < n; i++) cin >> arr[i];
    }
    void display() {
        for (int i = 0; i < n; i++) cout << arr[i] << " ";
        cout << endl;
    }
};

int main() {
    ArrayHandler<int> obj;
    cout << "Enter 3 ints: ";
    obj.input(3);
    obj.display();
    return 0;
}
```

```
PS C:\Users\HP> cd "d:\COLL
cpp -o ASS8CC } ; if ($?) {
Pair: Age, 20
PS D:\COLLEGE\oops\ass7> cd
+ ASS8DD.cpp -o ASS8DD } ;
Add: 16
Div: 5
PS D:\COLLEGE\oops\ass7> cd
+ ASS8EE.cpp -o ASS8EE } ;
Enter 3 ints: 4 5 6
4 5 6
PS D:\COLLEGE\oops\ass7>
```